

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I Hashini S / 192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

~~I understand that any violation of academic integrity rules will result in disciplinary action.~~

Signature of the student: Hahsini S

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



1. Learning Agent for Personalized Education System

1.1 Introduction

A Learning Agent is an intelligent AI system that observes a student's learning behavior, analyzes their performance, and automatically adjusts the learning material according to their needs.

In a personalized education platform, the agent continuously collects information such as:

- Quiz scores
- Assignment marks
- Time taken to answer questions
- Number of attempts
- Topics completed
- Incorrect answers
- Learning history
- Student interaction patterns

The agent uses this information to determine the student's knowledge level and learning progress and recommends suitable learning content.

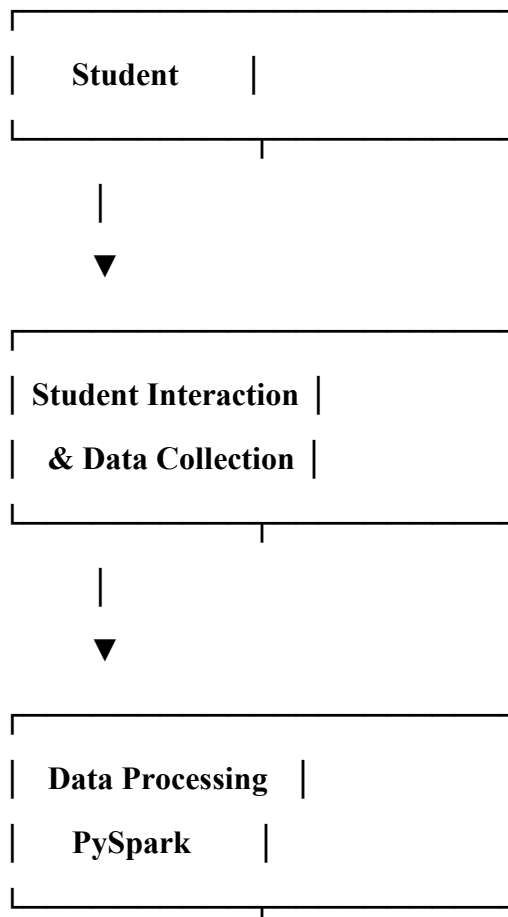
1.2 Objectives of the Learning Agent

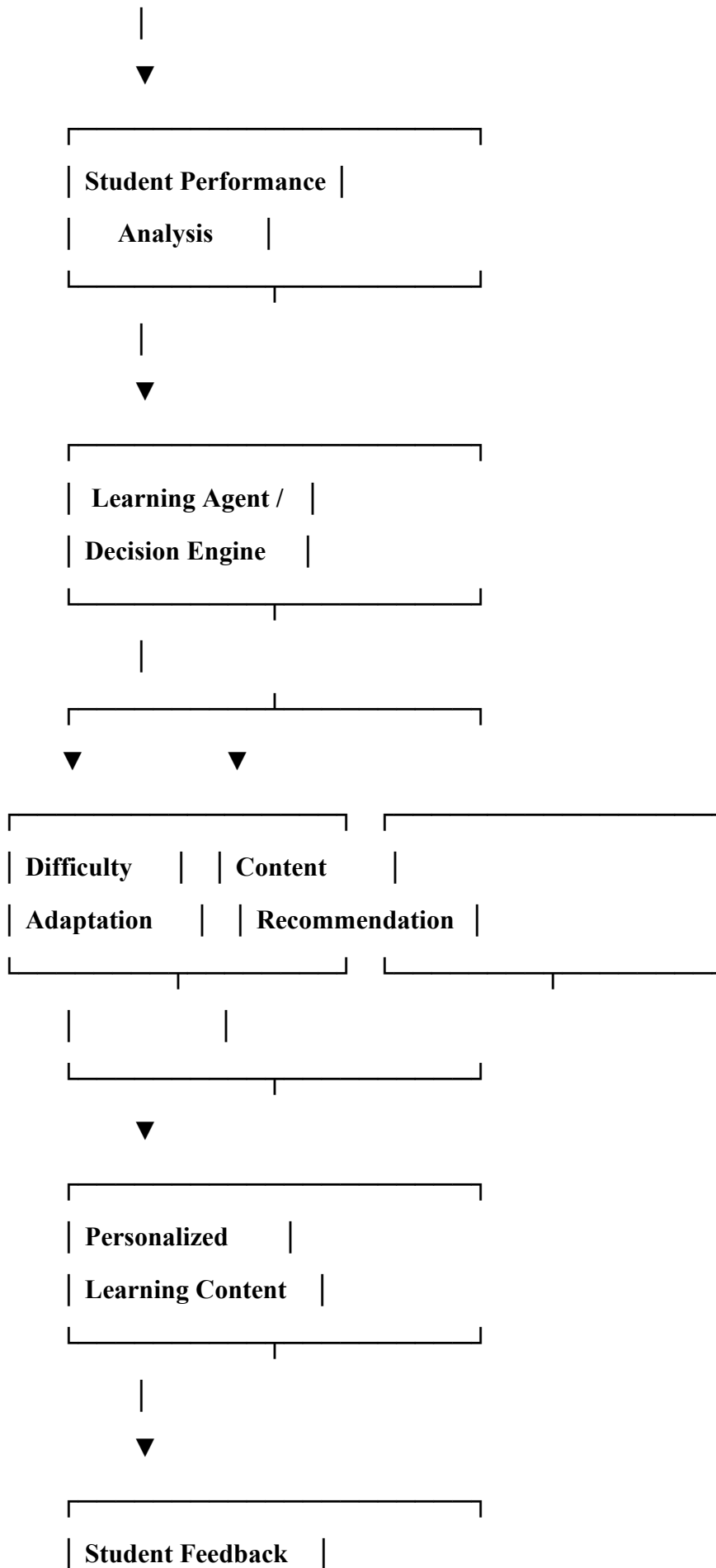
The main objectives are:

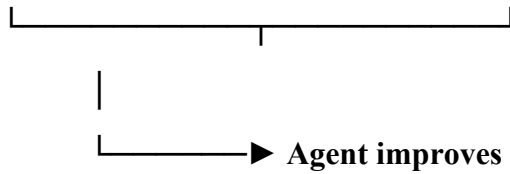
1. Personalize learning for each student.
2. Monitor student performance continuously.
3. Adapt content difficulty based on performance.
4. Provide immediate feedback.
5. Identify weak and strong topics.
6. Recommend appropriate learning resources.
7. Improve its decisions over time using feedback.
8. Process large-scale student data using PySpark.
9. Predict future learning performance.
10. Reduce learning gaps between students.

1.3 Architecture of the Learning Agent

The Learning Agent can be designed using the following major components:







1.4 Components of the Learning Agent

1. Student Data Collection

The first component collects information about student activities.

Example:

Student	Topic	Score	Attempts	Time
S01	Python	85	1	20 min
S02	Python	45	3	40 min
S03	Python	92	1	15 min

This data helps the agent understand individual learning behavior.

2. Data Processing Module

The collected data may be very large when thousands or millions of students use the platform.

PySpark can process this large-scale data efficiently using distributed computing.

Typical operations include:

- Filtering
- Mapping
- Grouping
- Aggregation
- Sorting
- Feature extraction

For example:

Student Data



PySpark



Clean Data

↓

Performance Features

↓

Learning Agent

3. Student Performance Analyzer

The agent calculates indicators such as:

Accuracy

$$Accuracy = \frac{Correct\ Answers}{Total\ Questions} \times 100$$

Average Score

$$Average\ Score = \frac{\sum Scores}{Number\ of\ Tests}$$

Other indicators include:

- Completion rate
- Attempt frequency
- Response time
- Topic-wise accuracy
- Improvement rate

1.5 Adaptive Content Difficulty

One of the most important features of the Learning Agent is automatic difficulty adjustment.

The agent changes the difficulty according to the student's performance.

Example

Suppose a student is learning Python.

Easy Question

↓

Student answers correctly

↓

Agent increases difficulty



Medium Question



Student answers correctly



Agent gives Hard Question

If the student repeatedly answers incorrectly:

Hard Question



Incorrect



Agent detects difficulty



Medium Question



Incorrect



Easy Question + Explanation

A simple rule can be:

Performance	Agent Action
> 80%	Increase difficulty
60–80%	Maintain difficulty
< 60%	Reduce difficulty
Repeated mistakes	Provide revision
Strong performance	Give advanced content

1.6 Feedback Mechanism

Feedback is essential for improving the Learning Agent.

After every learning activity, the agent receives feedback such as:

- Correct/incorrect answer
- Quiz score
- Student rating
- Time taken
- Number of attempts
- Whether recommended content was completed

The agent uses this feedback to improve future decisions.

Feedback Loop

Student



Learning Activity



Performance



Feedback



Agent Analysis



Updated Student Profile



Better Recommendation



New Learning Activity



Feedback



This creates a continuous learning cycle.

1.7 How the Agent Improves Over Time

Initially, the agent may have limited information about a student.

For example:

Student S01

Knowledge Level = Unknown

After several activities:

Python → 85%

Machine Learning → 72%

Statistics → 45%

The agent learns that the student is strong in Python but needs additional support in Statistics.

Therefore, it may recommend:

Statistics Basics



Practice Questions



Medium Questions



Assessment



Advanced Statistics

As more data is collected, the student profile becomes more accurate.

1.8 PySpark Integration

PySpark is used when the educational platform has a large amount of data.

For example, an online education platform may have:

10,00,000 Students



Millions of Quiz Attempts



Millions of Learning Records



PySpark Cluster



Distributed Processing

PySpark can perform:

- Large-scale data cleaning
- Student performance aggregation
- Feature engineering
- Recommendation data preparation
- Statistical analysis
- Machine learning using MLlib

Example:

```
from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \  
    .appName("LearningAgent") \  
    .getOrCreate()
```

```
df = spark.read.csv(  
    "student_data.csv",  
    header=True,  
    inferSchema=True  
)
```

```
df.groupBy("student_id") \  
    .avg("score") \  
    .show()
```

This calculates the average score for each student.

1.9 Example Scenario

Consider three students learning Machine Learning.

Student	Score	Agent Decision
A	92%	Advanced topics
B	74%	Medium-level practice
C	42%	Basic concepts + revision

Instead of giving the same content to everyone, the agent creates a personalized learning path.

Student A

Basics → Intermediate → Advanced → Projects

Student B

Basics → Intermediate → Practice → Assessment

Student C

Basics → Examples → Practice → Revision → Assessment

This demonstrates personalized education.

1.11 Advantages

The Learning Agent provides several benefits:

- Personalized learning experience
- Automatic difficulty adjustment
- Faster identification of weak topics
- Continuous performance monitoring
- Immediate feedback
- Better student engagement
- Scalable processing using PySpark
- Data-driven decision making
- Improved learning outcomes
- Supports large numbers of students

1.12 Conclusion

A Learning Agent provides an intelligent approach to personalized education. It continuously observes student activities, analyzes performance, adapts content difficulty, and provides suitable learning recommendations. Feedback allows the agent to improve its decisions over time. PySpark enables the system to process large volumes of student data efficiently using distributed computing.

Thus, the combination of AI + Learning Agent + Feedback + Adaptive Learning + PySpark can create a scalable and intelligent educational platform that provides each student with a personalized learning experience.